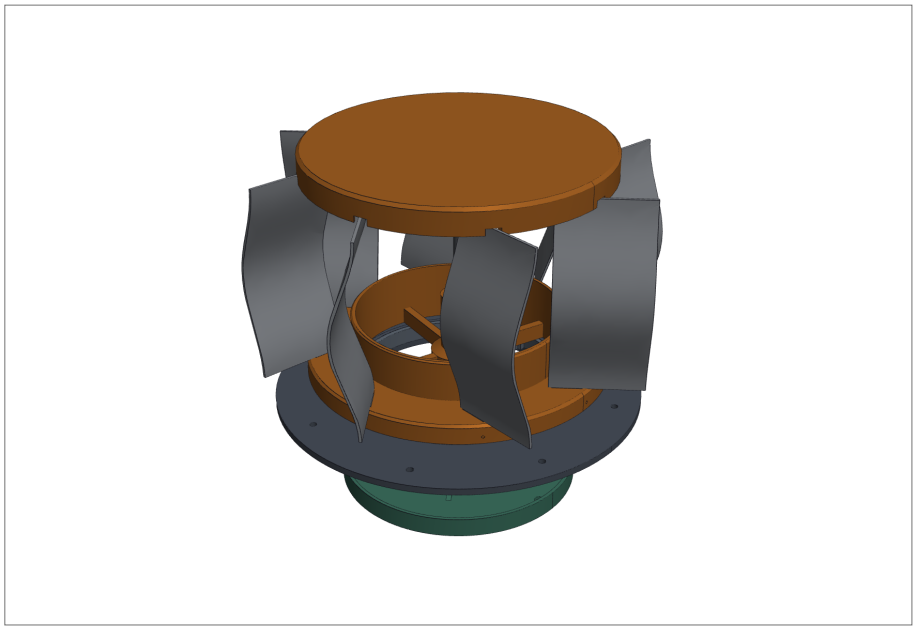
DOGHOUSE WHIRLYBIRD - CAD / FABRICATION PACKAGE

Rev DW-V1.1 - generated 2026-10-23 from the parametric build123d model (whirlybird/cad) and MASTER_PARAMETERS.json



COORDINATE SYSTEM / DATUMS

Z0 = finished roof surface at the rotor axis. +Z = rotor axis (roof to sky).
X/Y = plan. Blade 1 tab centre on +X axis; blades numbered CCW seen from above.
Datum A = shaft axis. Datum B = lower-hub top face (blade shoulder seat).
All STEP files share this origin; STL files are re-posed for printing.

Key result (from CAD)	Value
Installed height, roof to rotor top	165.2 mm (6.51 in) limit 190.5
Rotor OD incl. bowed blades	259.1 mm (10.20 in) target 266.7
Rotor OD at cap faces	245.4 mm
Hub OD upper / lower	188 / 188 mm
Cap separation (blade shoulders)	121.24 mm (bow 15 mm)
Blade chord angle from radial	35 deg, all same hand
Min blade-to-blade clearance	50.5 mm
Rotor-to-stationary axial gap	6.0 mm (labyrinth 2.0 radial)
Bearing spacing / shaft length	50 mm / 201.8 mm
Throat (roof opening)	127 mm (5 in hole saw)
Height margin to 190.5 limit	25.3 mm

DESIGN GATES (17/17 PASS) - see sheet 12 and reports/design_checks.json

#	Gate (automated on the CAD)	Result
1	No unintended solid interference anywhere in the assembly	PASS
2	Rotating parts never touch stationary parts (min clearance >= 1.5 mm, excl. bearing internal	PASS
3	Lower hub face to stationary collar/support axial gap = 6 mm	PASS
4	Blade-to-blade clearance (no collision incl. bow)	PASS
5	Blade-to-stationary clearance	PASS
6	Tab fits pocket with clearance on all sides and does not bottom out	PASS
7	Upper hub top face is one closed plane (no pocket breaks through)	PASS
8	Pocket walls to every exterior face >= 3.0 mm (top/bottom/OD)	PASS
9	Hex-key access to both hub set screws through blade gaps	PASS
10	Assembly sequence collision-free (blades drop into lower hub, upper hub drops over tabs)	PASS
11	Blades positively trapped once hubs are fixed (axial + radial)	PASS
12	Installed height <= 190.5 mm	PASS
13	Assembly CAD top equals reported installed height	PASS
14	Rotor clears roof surface by >= 6 mm (at modelled pitch)	PASS
15	Every printed part fits the X1E bed in print orientation	PASS
16	Shaft length fits user's 209.55 mm tube (if bearings accept it)	PASS
17	All STL meshes watertight, manifold, single body, volume within 1 %, on bed	PASS

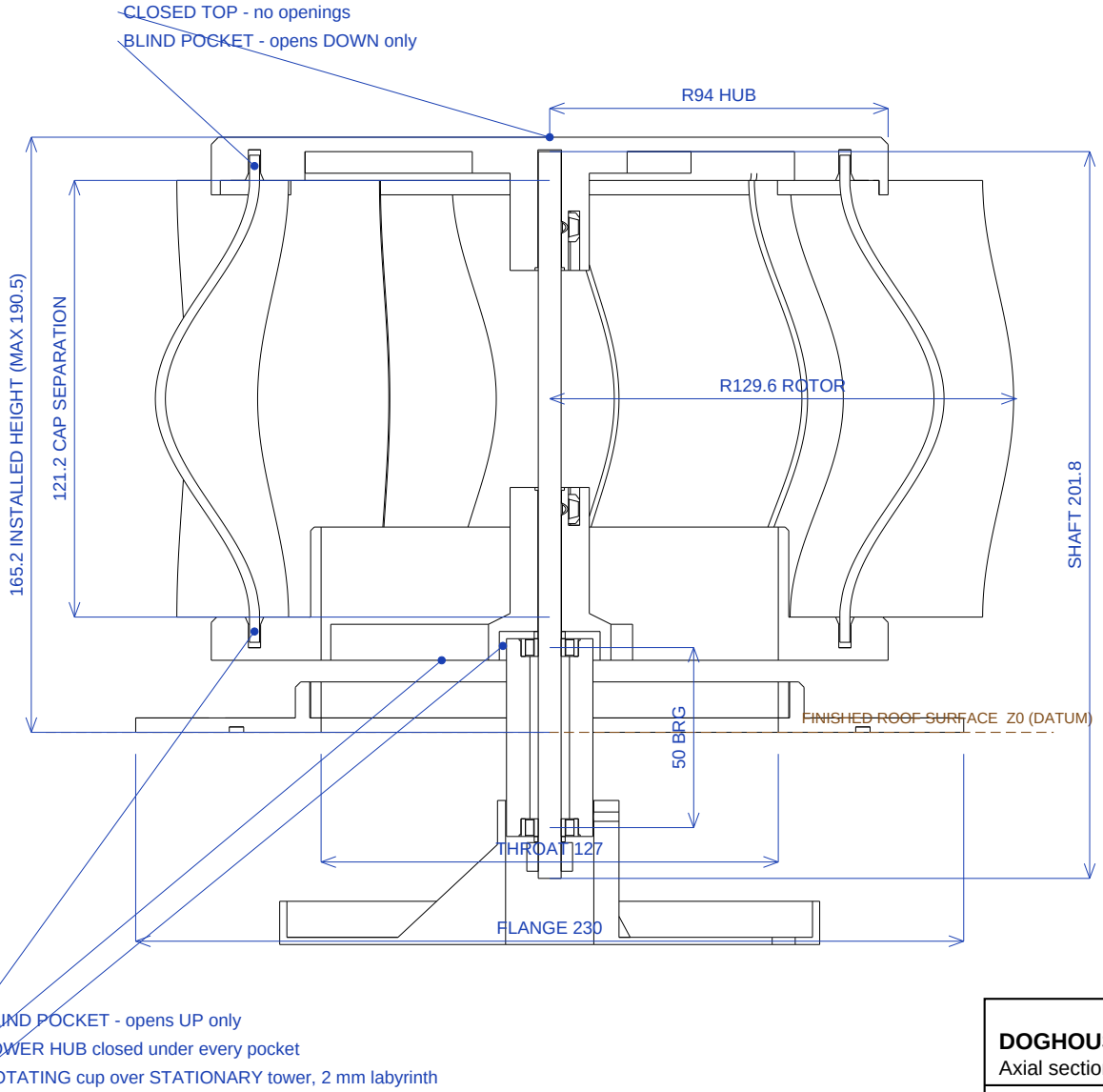
SHEET INDEX

1 Cover / datums / results
2 Axial section, height stack
3 Rotor vs stationary
4 Exploded assembly + order
5 True CAD views
6 Upper hub
7 Lower hub
8 Bearing support
9 Roof flash adapter
10 Blade 1:1 + forming
11 Coupons + tooling
12 Checks + calcs

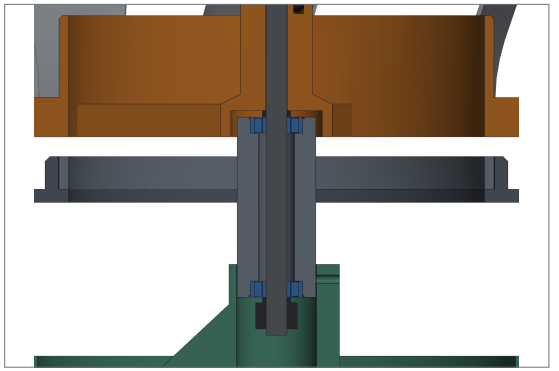
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Cover, datums, results

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

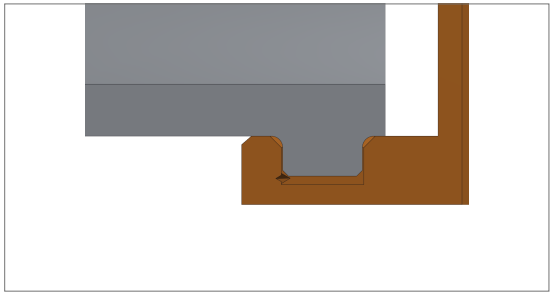
SHEET 2 - AXIAL HALF-SECTION (plane XZ, true CAD hidden-line projection, 1:2) + HEIGHT STACK



Bearing / labyrinth detail (render)



Lower pocket detail: tab, R2 roots, blind floor, weep



Height stack (Z, mm)	from	to	size
Roof to collar/support top	0	14.0	14.0
Rotor-to-stationary gap	14.0	20.0	6.0
Lower hub	20.0	32.0	12.0
Blade shoulders (cap sep.)	32.0	153.24	121.24
Upper hub	153.24	165.24	12.0
Bearing centres	-26.46	23.54	50.0
Shaft	-40.56	161.24	201.80

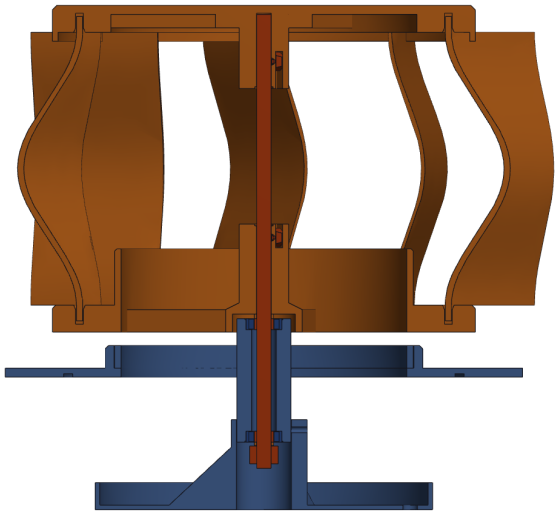
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Axial section, height stack

REV DW-V1.1 UNITS mm SCALE 1:2 2026-10-23

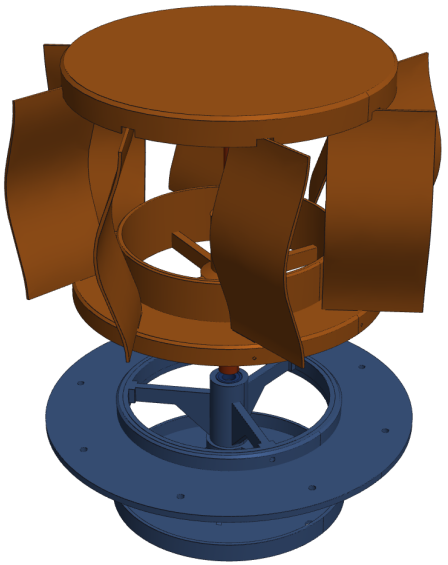
SHEET 2/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 3 - ROTOR vs STATIONARY: what turns, what is fixed to the roof, and how each attaches

SECTION - ORANGE rotating / BLUE stationary



Rotor (orange) lifted 70 mm off the stationary support (blue)



ROTATING part (orange)	Attaches to	By
Upper hub (closed top)	shaft (blind bore)	1x M4 set screw in nut slot -> shaft dimple
Lower hub (closed bottom)	shaft	1x M4 set screw in nut slot -> shaft dimple
8 blades	both hubs	tabs trapped in blind pockets (no fasteners)
Shaft 1/4 in	bearing INNER races	slip fit; lower hub bears on top inner race via washer
Top thrust washer	upper bearing inner race	loose on shaft, under lower-hub cup
Shaft collar + bottom washer	lower bearing inner race	collar set screw (uplift stop, 0.2 endplay)

STATIONARY part (blue)	Attaches to	By
Roof flash adapter	roof deck	8x #8 SS screws + sealant bead in groove
Bearing support (ring/spokes/tower)	adapter collar counterbore	rests on step; 3x M3 radial screws
2x R4-2RS bearing OUTER races	bearing support tower	pressed into seats (fit per coupon)
Rain tray (optional)	tower bottom (below roof)	slip-on standpipe + 1x M3 clamp screw

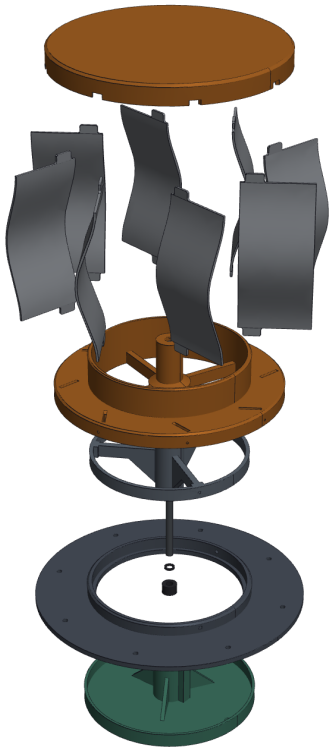
NO stationary part is fastened to either rotating hub. The only paths between rotor and stator are the two ball bearings. Minimum running clearances (from CAD): 6.0 mm lower hub to collar/support ring, 2.0 mm radial labyrinth (cup vs tower), 51 mm blade to blade, >13 mm blade to any stationary part.
Service: remove 3 M3 screws -> the whole rotor + bearing cartridge lifts out of the roof adapter.

DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Rotor vs stationary

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

 Source: build123d parametric model + MASTER_PARAMETERS.json

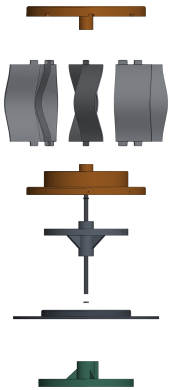
SHEET 4 - EXPLODED ASSEMBLY (true CAD) AND ASSEMBLY ORDER



ASSEMBLY ORDER (proved collision-free in CAD, gate 10):

1 Press both R4-2RS bearings into the support tower (top seat from above, bottom seat from below) - fit chosen from the fit coupon.
2 Bottom washer + 1/4 in shaft collar loosely on the shaft; slide shaft UP through both bearings from below.
3 Top thrust washer onto the shaft, on the upper inner race.
4 Lower hub down the shaft until its cup ceiling sits on the washer. Mark through the set-screw hole; spot-drill dimple (guide); M4 set screw + nut, threadlocker. Set collar for 0.2 mm endplay.
5 Drop each blade's lower tab straight DOWN into a lower-hub pocket (same orientation for all 8: bow outward, tab at inner edge).
6 Lower the upper hub straight DOWN over all 8 upper tabs (lead-in chamfers align them). Press lightly until every shoulder seats.
7 Mark, spot-drill the 2nd dimple, tighten upper set screw (thread-locker). Blades are now trapped: no per-blade hardware.
8 Spin by hand: no rub, no rattle. Static-balance (tape on hub rim).
9 Fasten roof adapter (sealant + 8 screws); drop the rotor cartridge into the collar; 3x M3 radial screws. Optional tray underneath.

Exploded offsets are along +Z only, matching the real assembly direction: blades enter the lower hub from ABOVE; the upper hub comes DOWN over the upper tabs. No part has to be threaded over an already-bent blade.



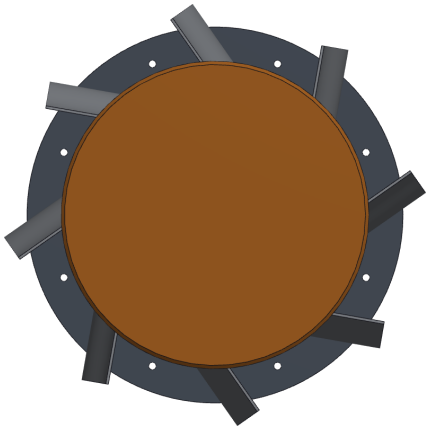
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Exploded assembly

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

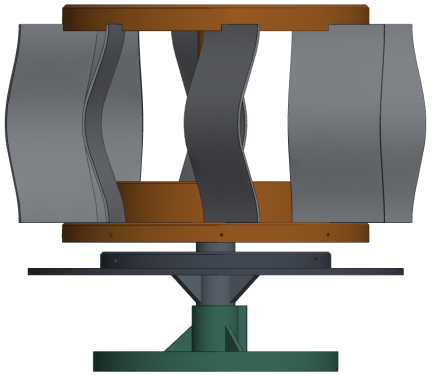
 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 5 - TRUE CAD-RENDERED VIEWS (tessellated build123d solids)

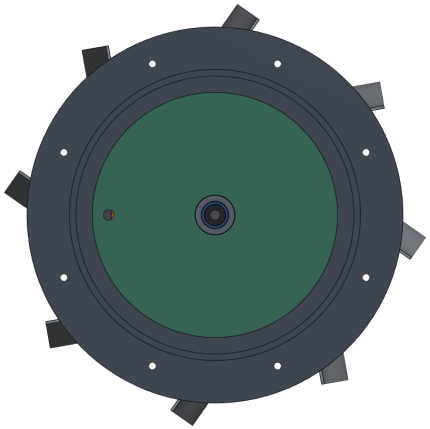
TOP



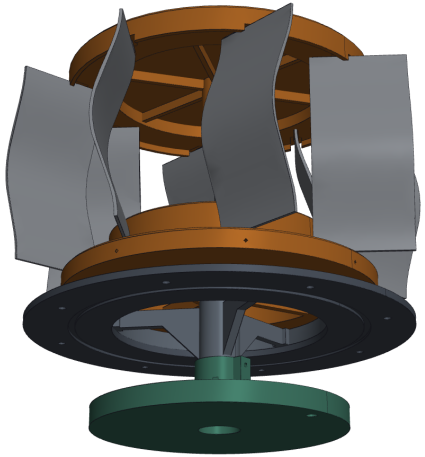
SIDE



UNDERSIDE



ISO FROM BELOW



DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR

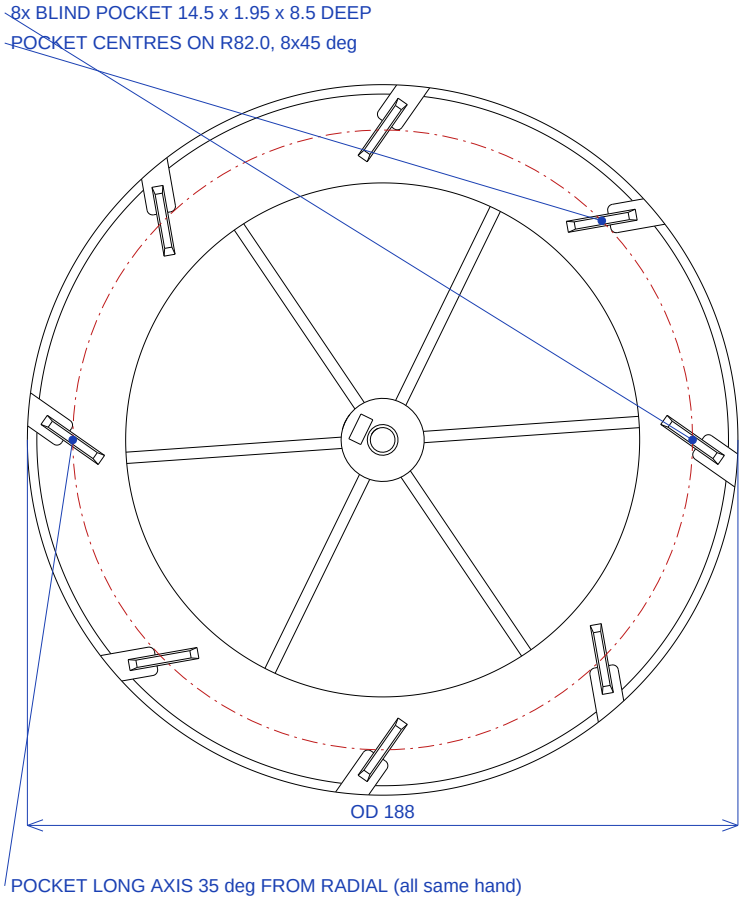
True CAD views

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

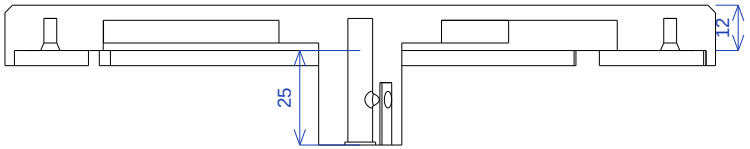
 Source: build123d parametric model + MASTER_PARAMETERS.json

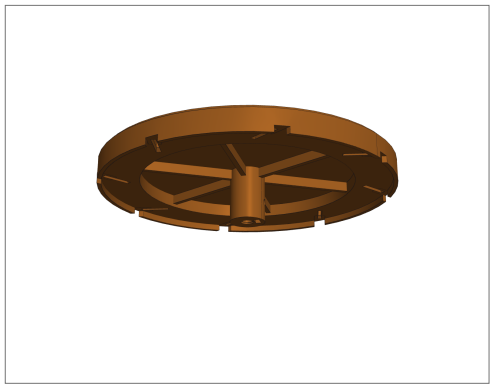
SHEET 6 - UPPER HUB CLOSED TOP (print TOP FACE DOWN)

PLAN FROM BELOW (pocket face) 1:2

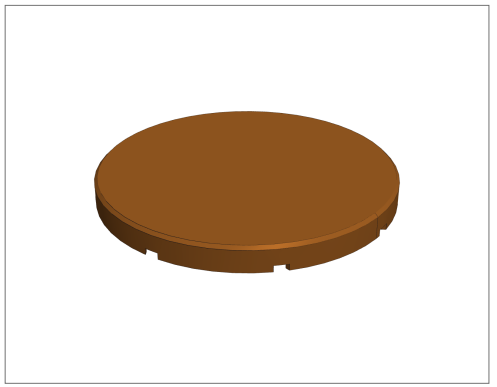


HALF SECTION XZ 1:2





pockets visibly BLIND



closed exterior TOP

Feature	Value
Pocket width (coupon-selected)	1.95 (try 1.85/1.95/2.05/2.15)
Pocket length / depth	14.5 / 8.5 (tab 14 x 7)
Mouth lead-in	ends 2.0 x45, sides 0.5
Intact wall over/under pocket	3.5 mm
Wall pocket->OD (mouth / deep)	3.48 / 5.44
Shaft bore	6.55 BLIND
Set screw	M4 x6 cup, nut slot 7.3x3.5, at 26 deg az
Set screw height	Z 140.24
Drip lip	2.5 x 4.0 notched at blades
Centre web / ribs	4.0 mm web, 6 ribs

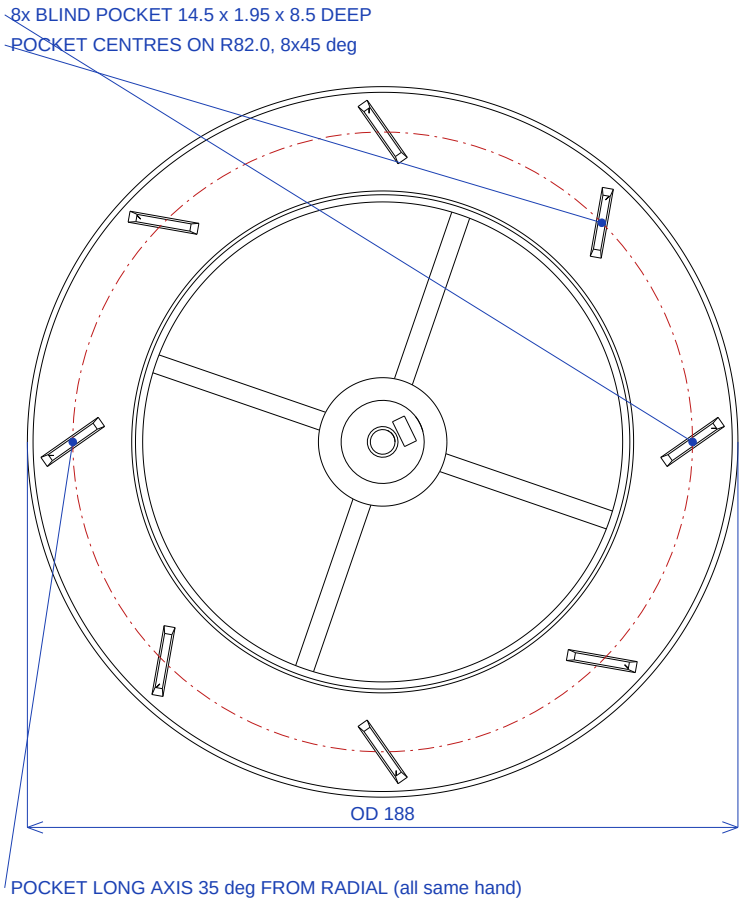
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Upper hub

REV DW-V1.1 UNITS mm SCALE 1:2 2026-10-23

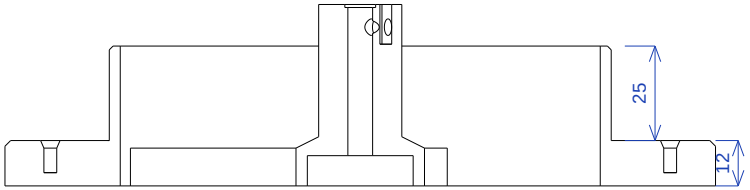
SHEET 6/12 Source: build123d parametric model + MASTER_PARAMETERS.json

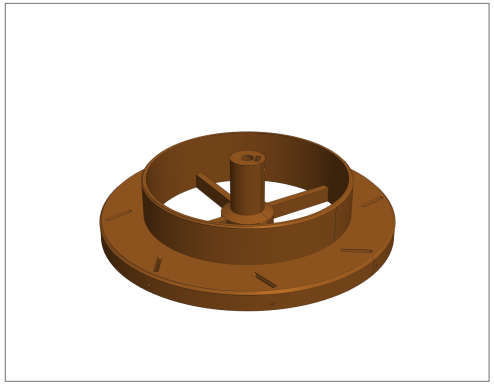
SHEET 7 - LOWER HUB CLOSED BOTTOM (print BOTTOM DOWN)

PLAN FROM ABOVE (pocket face) 1:2

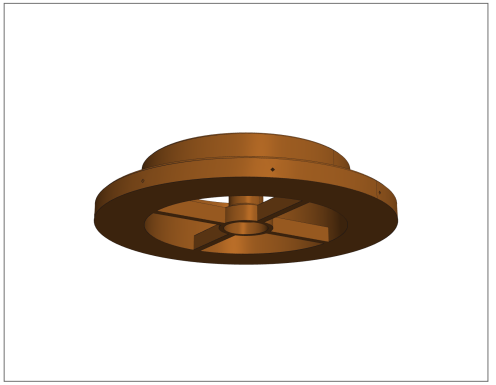


HALF SECTION XZ 1:2





pockets open UP only



closed exterior BOTTOM

Feature	Value
Pocket width (coupon-selected)	1.95 (try 1.85/1.95/2.05/2.15)
Pocket length / depth	14.5 / 8.5 (tab 14 x 7)
Mouth lead-in	ends 2.0 x45, sides 0.5
Intact wall over/under pocket	3.5 mm
Wall pocket->OD (mouth / deep)	3.48 / 5.44
Shaft bore	6.55 through boss
Set screw	M4 x6 cup, nut slot 7.3x3.5, at 26 deg az
Set screw height	Z 62.00
Weep channels	8x 1.6 diamond, pocket floor -> OD, 2 deg fall
Chimney ID / spokes	127 / 4 x 5 mm
Labyrinth cup	ID 28, ceiling Z 28.0

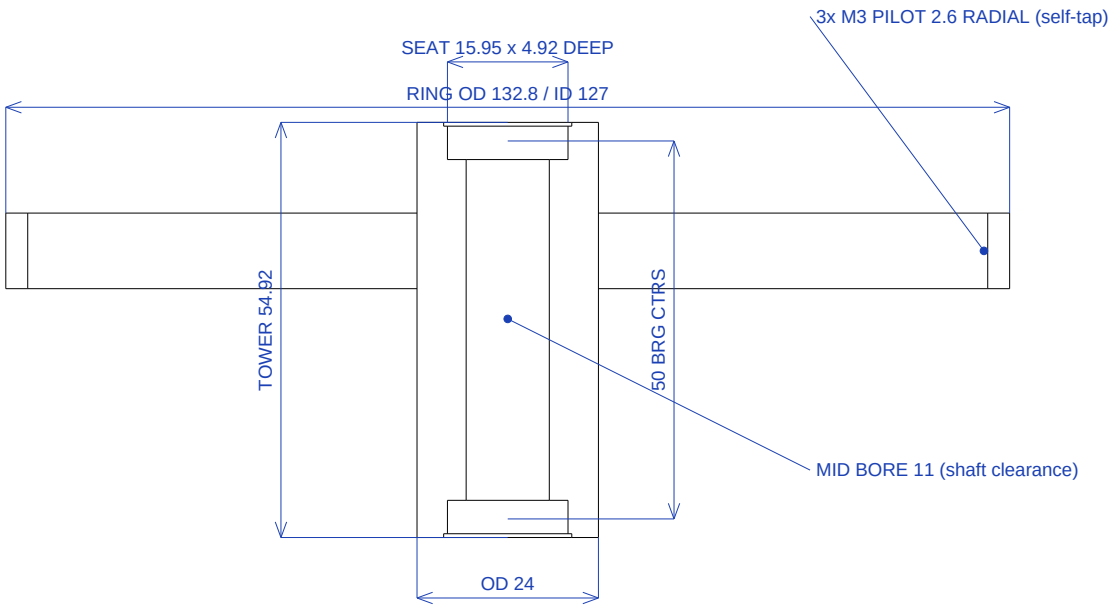
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Lower hub

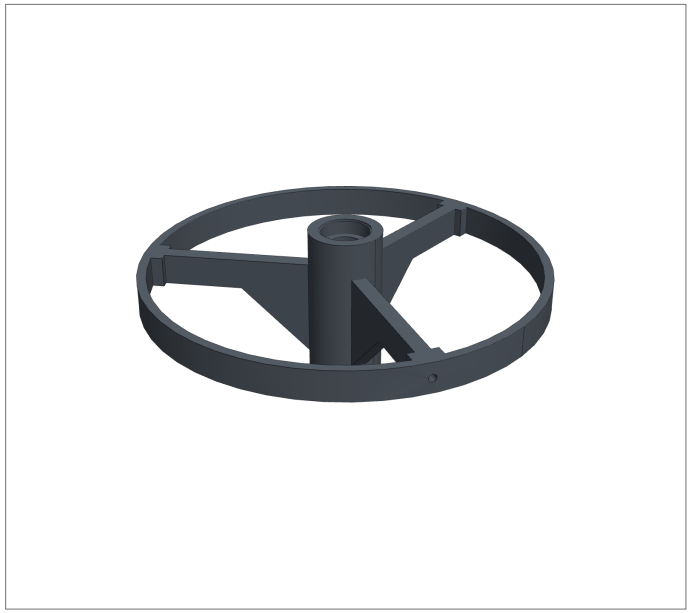
REV DW-V1.1 UNITS mm SCALE 1:2 2026-10-23

SHEET 7/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 8 - BEARING SUPPORT (STATIONARY) print TOWER-TOP DOWN, supports under ring/spokes

HALF SECTION XZ 1:1





Notes:
- Bearing OUTER races only are held here.
- Seat bores from fit coupon; do not print hard interference before testing.
- Gussets under each spoke stiffen the 43 mm tower below the ring.
- Ring top flush with collar top: 6 mm below the rotating lower hub.

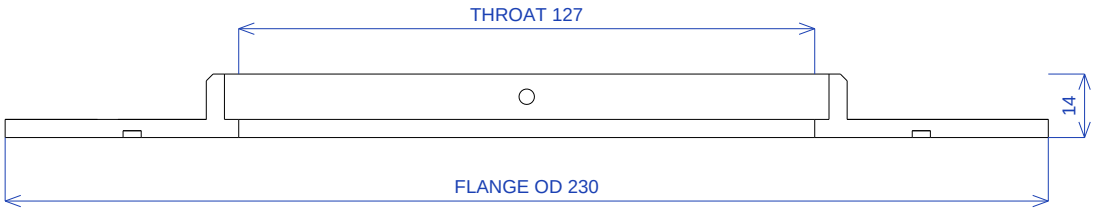
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Bearing support

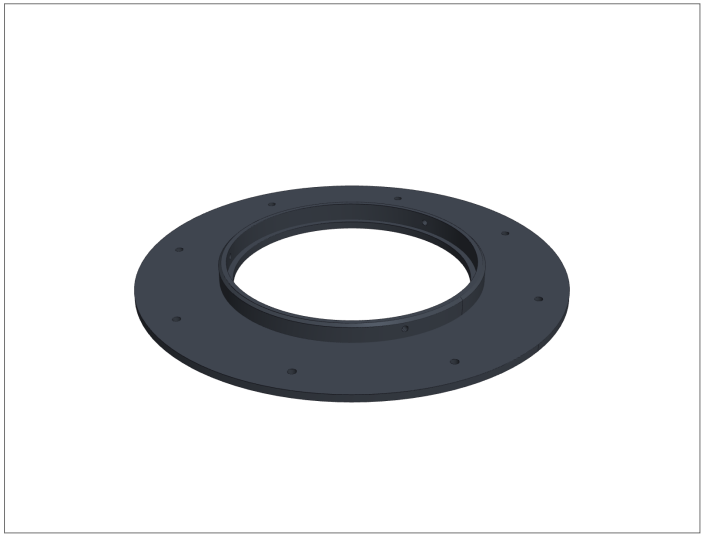
REV DW-V1.1 UNITS mm SCALE 1:1 2026-10-23

SHEET 8/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 9 - ROOF FLASH ADAPTER (STATIONARY) pitch modelled = 0.0 deg

HALF SECTION XZ 1:1.67





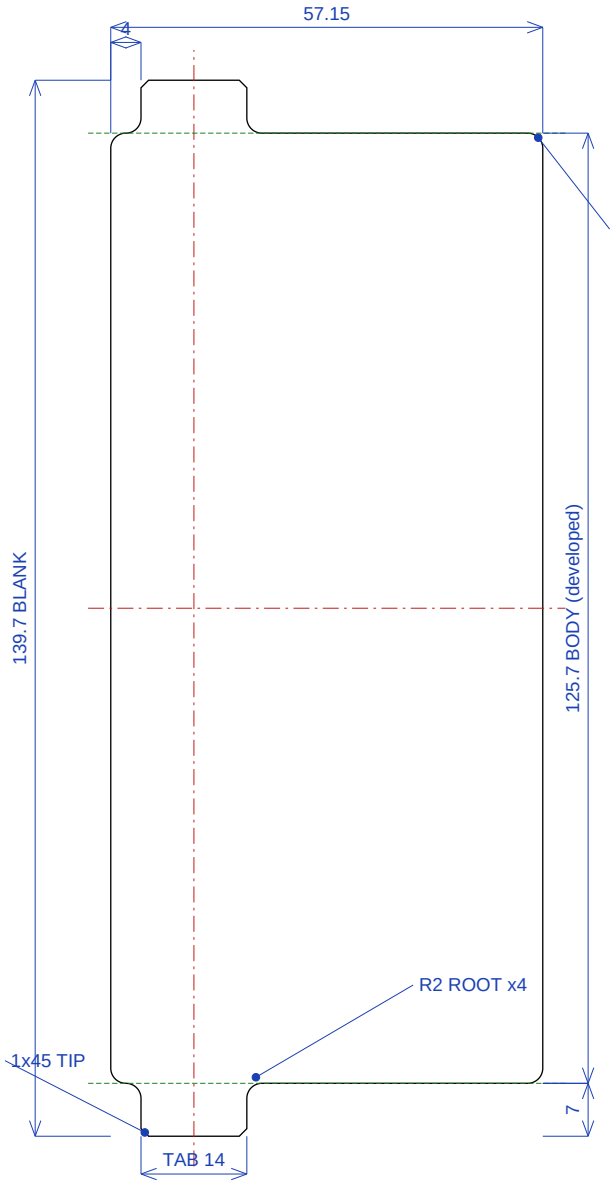
Roof pitch file	deg	Collar top Z	Installed height	<=190.5
roof_flash_adapter_pitch_0in12.stl	0.0	14.0	165.2	YES
roof_flash_adapter_pitch_2in12.stl	9.46	23.1	174.3	YES
roof_flash_adapter_pitch_3in12.stl	14.04	28.6	179.9	YES
roof_flash_adapter_pitch_4in12.stl	18.43	34.2	185.4	YES
roof_flash_adapter_pitch_5in12.stl	22.62	39.8	191.0	NO

ROOF CUT-OUT: 127 mm (5 in) hole saw, centred. Seal the plywood cut edge with exterior paint.
FASTEN: 8x #8 x 3/4 in stainless pan-head wood screws with EPDM washers on a 200 mm circle.
SEAL: continuous bead of butyl / polyurethane in the 4 x 1.5 mm groove (R87),
plus a fillet bead round the flange edge on the UPHILL half. On shingles, slip the uphill half under the course above (no sealant needed there) - classic step-flash logic.
PITCH: real pitch unknown. Pitched files tilt the flange; the collar stays vertical. Above ~5/12 the 190.5 mm limit is exceeded because the rotor must lift to clear the uphill roof.
SUPPORT: bearing-support ring drops into the counterbore and rests on the step; 3x M3 screws radial.

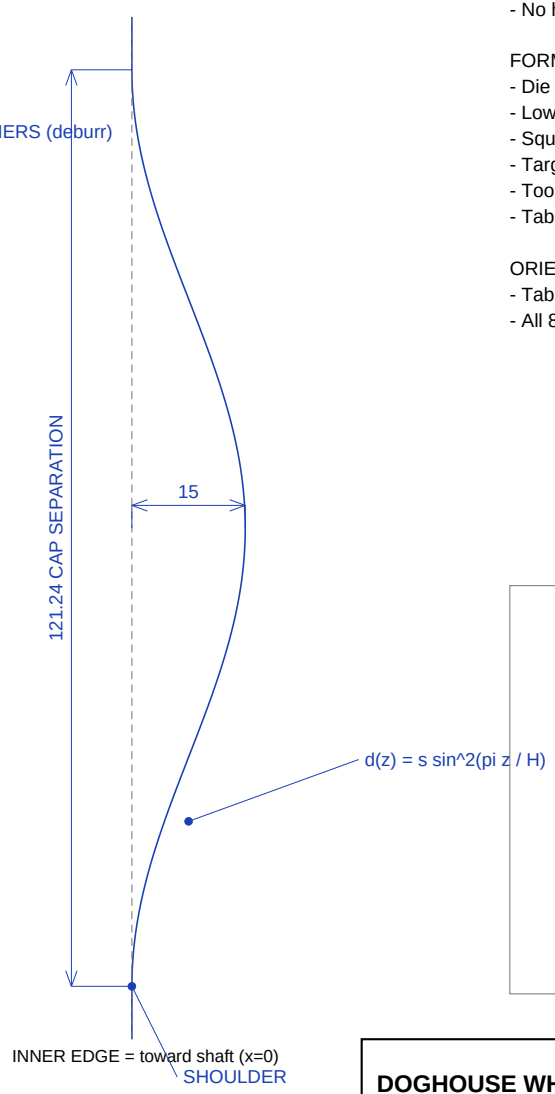
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Roof flash adapter

REV DW-V1.1 UNITS mm SCALE 1:1.67 2026-10-23

SHEET 10 - BLADE: FLAT PATTERN 1:1, FORMED PROFILE 1:1, FORMING



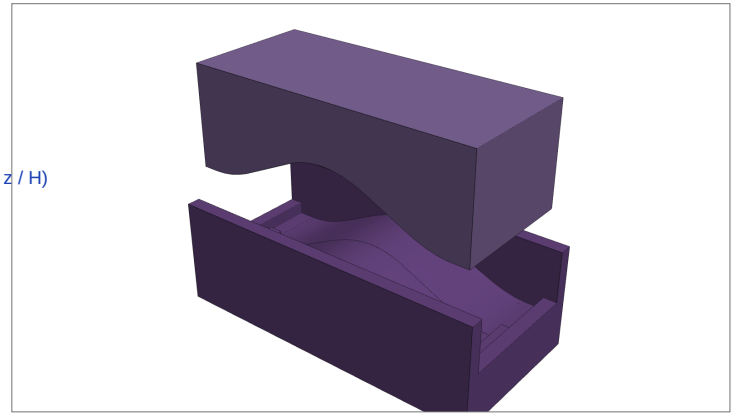
FORMED SIDE PROFILE 1:1 (lay blade on it)



CUT (gate C: ONE blade first)
- Rip stock 3.75 in -> 57.15 mm; crosscut 139.7 mm; file to the line; kerf on the WASTE side.
- Notch tabs: saw/snips to 2 mm short of the shoulder, file R2 roots (no sharp inside corners).
- Break all edges; R2 outer corners; 1x45 tab tips.
- No holes in the blade.

FORM (printed press die, bench vise)
- Die sagitta 19.5 mm = 15 x springback 1.30.
- Lower die keys only accept the blank tab-side correct.
- Squeeze slowly until faces meet; release; measure.
- Target bow 15 +/-1.5 mm; shoulders 121.2 +/-0.5 apart.
- Too flat -> raise springback factor, reprint die.
- Tabs must stay flat, in line, and parallel.

ORIENTATION IN ROTOR
- Tab toward the shaft-side edge; bow bulges OUTWARD.
- All 8 identical; all same hand.

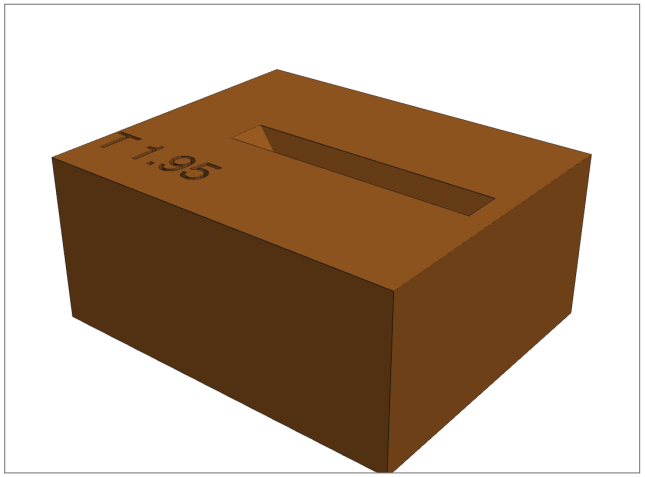


DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Blade

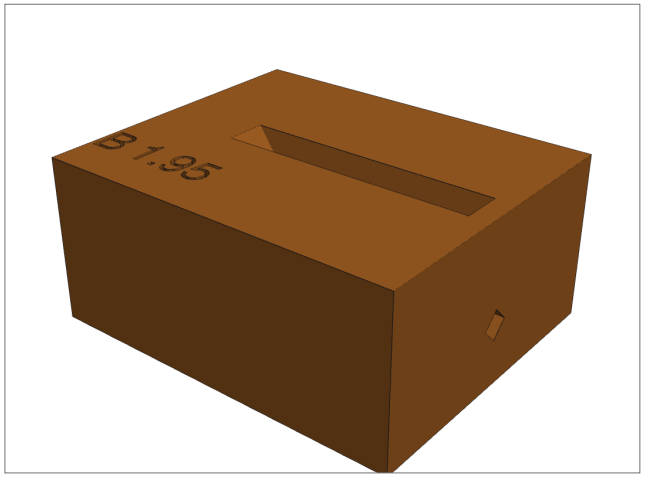
REV DW-V1.1 UNITS mm SCALE 1:1 (blade) 2026-10-23

SHEET 10/12 Source: build123d parametric model + MASTER_PARAMETERS.json

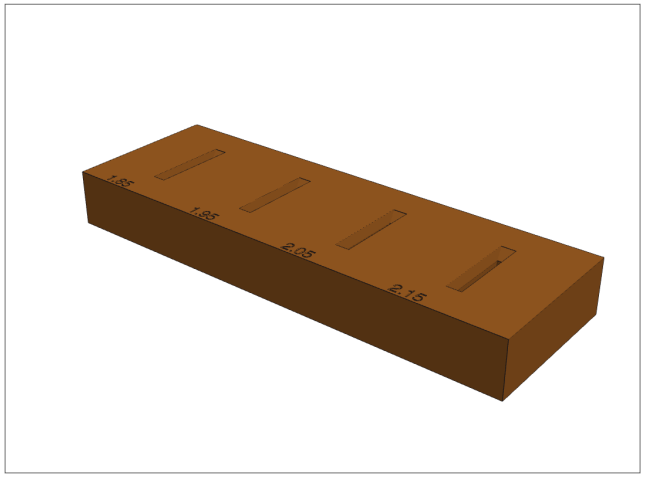
SHEET 11 - PRINT THESE FIRST: FIT COUPONS AND TOOLING



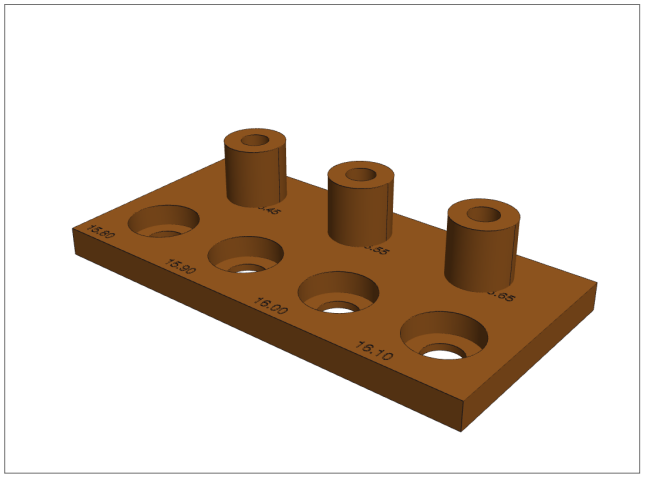
single_blade_socket_test_top (closed top, opens down)



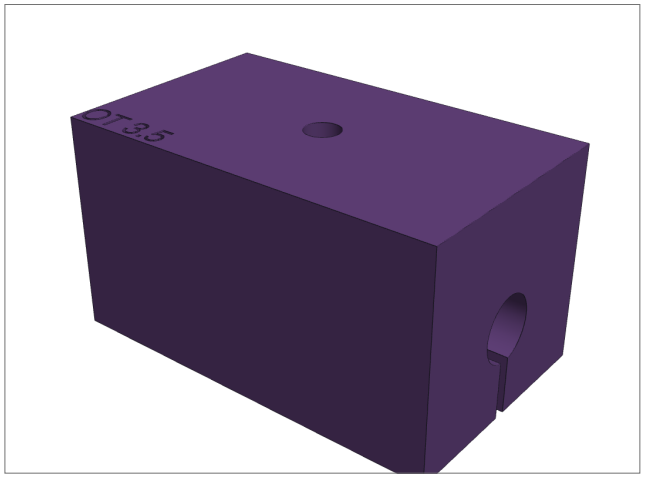
single_blade_socket_test_bottom (opens up, weep)



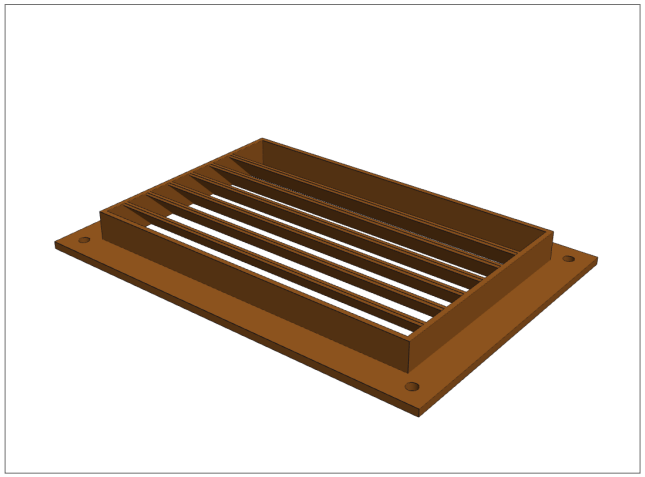
slot_clearance_coupon 1.85 / 1.95 / 2.05 / 2.15



bearing seats 15.80-16.10, shaft bores 6.45-6.65



shaft_dimple_drill_guide (3.5 mm spot)



inlet_vent_louver x2 (screened, 45 deg)

PASS CRITERIA (gate B): 1.6 mm aluminium tab slides in by hand, does not rock more than ~0.3 mm at the tip, no whitening/cracking or layer splitting when wiggled, cannot be pushed through any outer face. Choose the narrowest slot that passes, enter it in MASTER_PARAMETERS.json (pocket.width) and rebuild. Bearings: firm thumb-press, no rattle. Shaft bore: slides with light force, no wobble.

DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Coupons + tooling

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

SHEET 11/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 12 - AUTOMATED CAD CHECKS, CALCULATIONS AND OPEN ITEMS

Calculation	Value	Comment
Blade mass (1.6 Al)	31.9 g	set of 8: 255 g
Centrifugal load @1000 rpm	35.4 N/blade	CG radius 101 mm
Per tab	17.7 N	bearing stress 1.6 MPa on 1.6x7 tab
Proof pull target	50 N per tab	= 2.8x the 1000 rpm load; keep
Pocket end-wall shear-out	> 1 kN est.	4 mm wall x 8.5 deep x 2 shear planes x ~15 MPa
Hub hoop stress @1000 rpm	< 0.1 MPa	negligible
Thermal (Al vs SS shaft, 60 K)	0.04 mm	cap separation change, << 7 mm engagement
Wind uplift on upper hub (25 m/s)	~5-10 N	resisted by set screw in dimple (positive)

OPEN ITEMS / NEEDS PHYSICAL VALIDATION

- Roof pitch is unknown: CAD built at roof_pitch_deg=0.0. Adapters for 0, 2/12, 3/12, 4/12 and 5/12 are exported; measure and rebuild.
- Roof deck thickness assumed 15 mm (only affects tower clearance below the deck).
- Pocket width 1.95 mm is a placeholder until the 1.85/1.95/2.05/2.15 coupon is tested with your actual aluminium.
- Bearing seat bore 15.95 mm and hub shaft bore 6.55 mm are placeholders until the bearing/shaft fit coupon is tested.
- Bearing bore not verified: confirm the R4-2RS markings and that the chosen shaft slips through both inner races without force.
- Forming-die springback factor 1.30 is a first guess; the alloy/temper of the 1.6 mm sheet is unknown.
- Cap separation 121.24 mm assumes the formed bow equals 15.0 mm; actual separation is set at assembly by seating the upper hub on the blade shoulders.
- Airflow, rpm and rain ingress are not predicted by this CAD; they must be measured (spec gates E and F).

MESH VERIFICATION (trimesh)

STL	watertight	bodies	dV %
upper_hub_closed_top.stl	yes	1	-0.048
lower_hub_closed_bottom.stl	yes	1	-0.08
thrust_washer_top.stl	yes	1	-0.06
thrust_washer_bottom.stl	yes	1	-0.06
bearing_support_stationary.stl	yes	1	-0.065
roof_flash_adapter.stl	yes	1	-0.06
rain_baffle_if_needed.stl	yes	1	-0.047
blade_forming_die_lower.stl	yes	1	0.028
blade_forming_die_upper.stl	yes	1	-0.109
single_blade_socket_test_top.stl	yes	1	0.0
single_blade_socket_test_bottom.st	yes	1	0.0
slot_clearance_coupon_1p85_1p95_2p	yes	1	0.0
bearing_and_shaft_fit_coupon.stl	yes	1	0.002
shaft_dimple_drill_guide.stl	yes	1	0.004
inlet_vent_louver.stl	yes	1	0.0
lower_hub_SOLID_DISC_ALTERNATE_not	yes	1	-0.062
roof_flash_adapter_pitch_0in12.stl	yes	1	-0.06
roof_flash_adapter_pitch_2in12.stl	yes	1	-0.037
roof_flash_adapter_pitch_3in12.stl	yes	1	-0.036
roof_flash_adapter_pitch_4in12.stl	yes	1	-0.027
roof_flash_adapter_pitch_5in12.stl	yes	1	-0.027

DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Checks + calcs

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

 Source: build123d parametric model + MASTER_PARAMETERS.json